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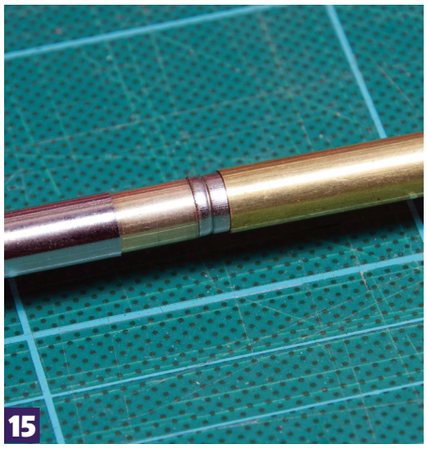
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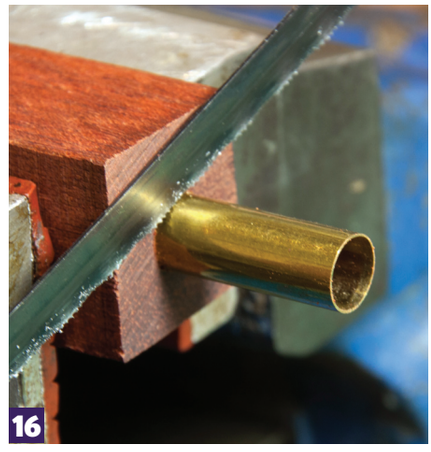
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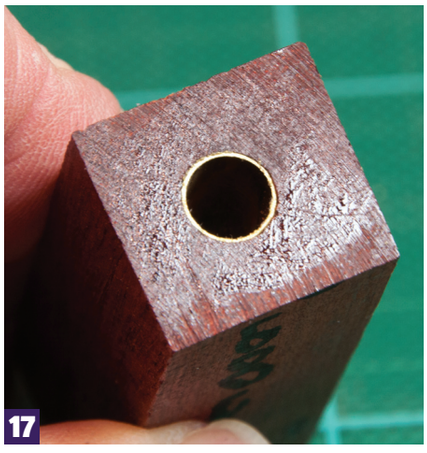
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11 Should adjustments to length be needed, to achieve the correct tip extension once the components are assembled, trim down with a thin parting tool

12 With the refill and spring installed and the mechanism screwed into place, it should be possible to extend the tip of the refill smoothly by twisting the mechanism and retracting it against the spring pressure without any sticking. Now is the time to make any adjustments required to achieve this

13 Cut the bloodwood blank to length and bore it out for the cap end of the pen. Drill in stages to first accommodate the brass tube and then the body of the mechanism. The exact sizes and depths will be determined by measuring the components you are using

14 Square off the ends of the blank true to the bore, using a jig on the disk sander or with a barrel trimmer. In this instance, I did this before installing the brass tube. You may need to make suitably sized sleeves for the jig to ensure accuracy

15 The tube from the Flat Top kit should be a good fit over the end of the mechanism, providing just enough grip to turn it without slipping. If it is too loose, slightly compress the very end of the mechanism to a slight oval shape with a pair of pliers to resolve this. The tube will not slide all the way onto the mechanism and it does not need to. Slide it as far as it will go and then pull it back to about 5–6mm for clearance

16 Slide the drilled wooden blank over the tube and mechanism until it is firmly seated against the Ebonite lower body. Mark the length of the protruding tube with a scriber or hacksaw, then remove the blank and tube and cut the tube to the marked length

17 Clean up the end of the tube with files or abrasive cloth and glue it into what will become the clip end of the blank. Using epoxy, make a good strong joint that will stand up to the twisting action of operating the pen. Of the alternatives, cyanoacrylate (CA) is not the best adhesive for wood-to-metal joints, while polyurethane glue, which might otherwise be a good choice, risks expanding into the part of the bore that does not contain the tube

18 With the refill and spring removed, mount the Ebonite section of the pen in a collet chuck with a suitably sized collet fitted to engage with the body of the mechanism, while supporting the tip end by a revolving centre in the tailstock. The Axminster Trade Series Precision Pro Lathe (that I review on pg. 94) is ideally suited for this. Do not overtighten the tailstock quill or you will flare the tip

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